

***Pluskin*: Interactive Machine-Embroidered Plush Skin for Creative and Affective Robot-Human Interaction**

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Fig. 1. The soft and fluffy texture of *Pluskin* naturally invites users to touch and interact with the robot dog.

We present *Pluskin*, a machine-embroidered plush skin that transforms the surface of a companion robot dog into a soft tactile sensing interface. Fabricated via digital loop-pile embroidery, *Pluskin* provides a scalable sensing solution that integrates the softness and familiarity of plush materials with textile-based sensing. We implemented a 4×3 sensing array on the dorsal surface of a Peto Bittle robotic dog to validate the system. Using a Support Vector Machine (SVM) classifier on extracted spatiotemporal voltage features, the system identifies five natural touch gestures and maps them to corresponding embodied robot behaviors. By integrating plush materiality, tactile sensing, and embodied robot feedback, this work demonstrates a feasible approach toward a more natural and affective human-robot interaction interface.

CCS Concepts: • **Human-centered computing** → **Ubiquitous computing**.

Additional Key Words and Phrases: Plush skin; Affective interaction; E-textiles; Companion robot; Gesture recognition; Human-robot interaction

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1 Introduction

As companion and social robots transition from task-oriented tools to daily life partners, the research focus has shifted from mere functional utility to the cultivation of deep human-robot relationships [6]. Within this paradigm, touch is not merely a data input but a vital channel for establishing intimacy, trust, and embodied empathy [3, 9, 10]. However, many existing robotic interfaces still rely on rigid materials such as plastic or silicone and on imperative input methods such as buttons, screens, or voice commands, which are poorly aligned with the subtle and relational qualities of natural social touch and affective gestures [10].

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Recent advancements in digital textile fabrication, such as e-textiles and machine knitting, have introduced flexible alternatives to rigid sensing interfaces [1, 2, 5]. Nevertheless, predominant paradigms in tactile sensing still prioritize thin-layered, flat architectures [7, 8, 11]. While these approaches are effective for pressure sensing and continuous input, they do not fully capture the soft, voluminous, and inviting qualities that are particularly relevant to companion robots. Plush materials are inherently inviting and evoke strong pet-like associations; they have historically been treated as passive decorative layers rather than active sensing media [4, 6]. This leaves a clear gap in leveraging the volumetric thickness and fibrous morphology of plush textures for functional tactile transduction.

To bridge this gap, we present *Pluskin*, a sensory plush skin realized through digital loop-pile embroidery. Unlike conventional film sensors or flat textile sensors, *Pluskin* exploits the inherent physical compliance and contact piezoresistive effect of embroidered loop-pile yarn structures. This approach transforms the traditionally decorative plush surface into a tactile interface capable of capturing the subtle spatiotemporal dynamics of human touch, as seen in Figure 1.

We validated the system by integrating *Pluskin* onto the dorsal surface of a Petoï Bittle quadruped robot ¹. By leveraging the rich spatiotemporal dynamics of the embroidered skin to capture voltage variations, an SVM-based classifier accurately identifies five natural social gestures, ranging from rhythmic stroking to impulsive patting. Mapping these signals to embodied behavioral feedback achieves a closed-loop interaction that transcends conventional command-based inputs. Through this system, we explore how plush materiality, tactile sensing, and embodied robot feedback can be combined to support more natural and affective human-robot interaction.

2 System Concept

The core concept of *Pluskin* is to **transform plush from a passive covering into an active tactile interface for human-robot interaction**. The system realizes a shift from command-based input to affective interaction through the following three-layer architecture:

(1) **Digitally Embroidered Plush Material and Sensing Layer:** By exploiting the thickness features inherent in digital loop-pile embroidery, this layer emulates the volumetric compliance of biological skin and captures physical deformations via the piezoresistive effect of conductive yarns.

(2) **Spatiotemporal Gesture Recognition Layer:** The system interprets tactile signals as continuous spatiotemporal patterns. Through machine learning, it extracts voltage signal features from the compressed plush surface, enabling recognition of natural gestures such as stroking or patting.

(3) **A Closed-loop Robot Dog Response Layer:** Recognized tactile cues are mapped onto the robot’s embodied behavioral feedback, e.g., tickling or sitting. This establishes a closed action-reaction loop on the Petoï Bittle robotic dog, evoking the intuitive experience of human-pet companionship.

Together, these three layers allow *Pluskin* to function as both a soft surface material and an interactive sensing interface.

3 Technical Implementation of the Plush Sensing Skin

3.1 Design and Scalable Fabrication of the Plush Sensing Interface

We developed a plush interface with tactile sensing functionality based on a piezoresistive mechanism (see Figure 2c). When force is applied to the surface, the conductive fiber network reorganizes, resulting in a measurable change in

¹<https://www.petoi.com/pages/bittle-smart-robot-dog-model-overview>

resistance—and consequently, voltage output. This mechanism transforms the passive plush surface into an active, interactive interface.

Structurally, as shown in Figure 2a, the sensor consists of three layers: (1) The plush layer, which is the core interactive surface, uses 120D/3 silver fiber-polyester blend conductive yarns. After testing various pile structures, we selected *loop pile* for its superior softness, structural stability, and elastic recovery under repeated contact. (2) The base layer is made of polyester fabric, providing structural support for the embroidered circuitry. (3) The adhesive layer uses eco-friendly glue to enhance overall stability and ensure reliability during use.

A key feature of our approach is its scalable, large-area digital fabrication process. Each sensing unit measures 20 mm × 20 mm (see Figure 2b), the sensor layout can be designed in Wilcom computer graphics software and then precisely embroidered onto the base fabric using YS-1201CT embroidery machine. Compared to manual fabrication, this method offers better repeatability, customizability, and potential for scaling.

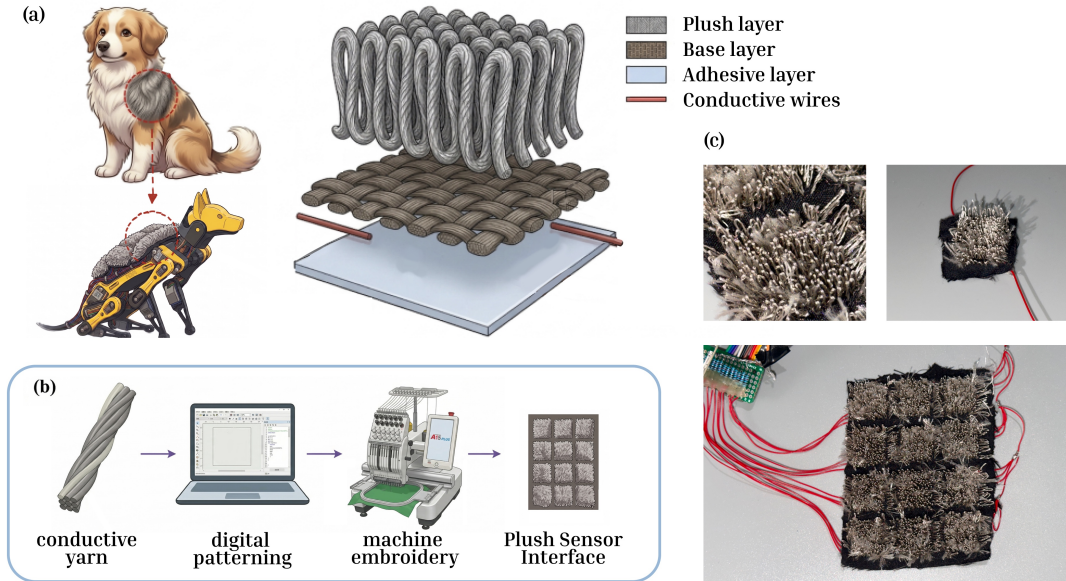


Fig. 2. (a) Exploded diagram of the sensor structure; (b) Fabrication process; (c) Close-up photograph of the sensor.

3.2 Hardware and Data Acquisition System

Based on the above process, we designed a 4×3 array of 12 sensing channels to cover the approximately 10 cm × 15 cm interactive area on the back of a robot dog (see Figure 3a). Each sensing unit corresponds to a local tactile region, enabling detection of spatial and temporal patterns of touch.

An Arduino Mega 2560 serves as the central controller, reading analog voltage signals from each channel. Each channel is numbered according to its position, enabling multi-channel continuous recording. Compared to single-point sensing, this array-based approach preserves spatial and temporal features of tactile input.

To support data collection and model training, we incorporated an SD card module to record raw sensor data locally. This provides a traceable dataset for machine learning.

4 Natural Tactile Gesture Recognition via Spatiotemporal Features and Machine Learning

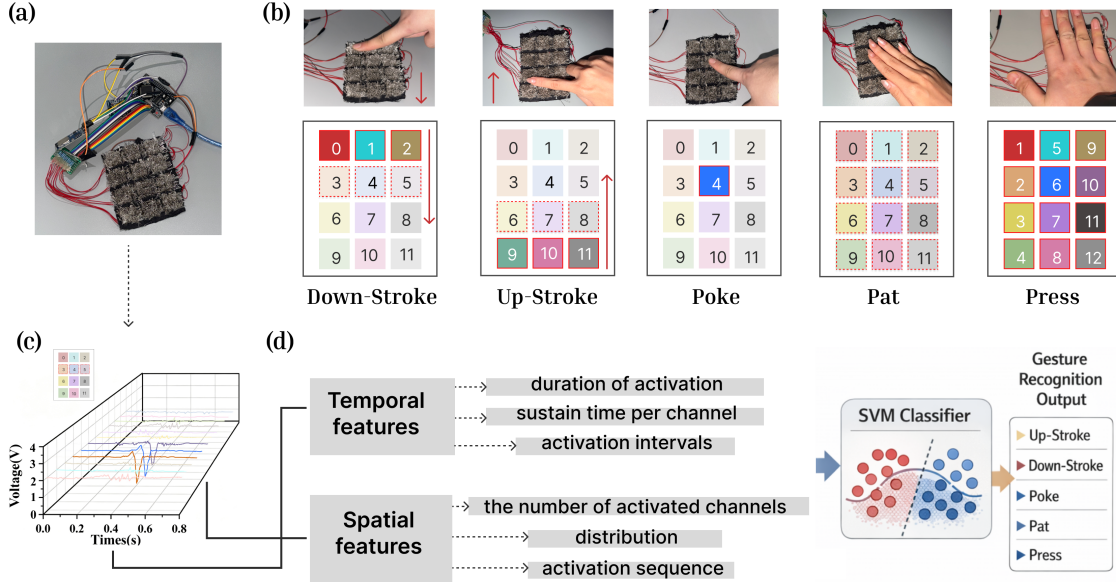


Fig. 3. Natural tactile gesture recognition based on the 4×3 plush sensing array. (a) Plush sensing skin and data acquisition hardware; (b) Representative activation patterns of five natural tactile gestures on the 4×3 array; (c) Synchronized multi-channel voltage responses during tactile interaction; (d) Machine learning pipeline, where temporal and spatial features are extracted from the array signals and fed into an SVM classifier for gesture recognition.

To enable recognition of natural tactile input on the plush skin, we defined five common gestures based on typical human-pet interactions: **up-stroke**, **down-stroke**, **poke**, **pat**, and **press**. These gestures not only align with intuitive touch behavior but also produce distinct spatiotemporal response patterns on the embroidered array (see Figure 3b). Definitions and characteristic features of each gesture are summarized in Table 1.

During signal processing, the system first detects valid activation: when a channel's voltage drops by 70% relative to its baseline, it is marked as triggered (see Figure 3c). We then extract two types of features (see Figure 3d): 1) Temporal features: including duration of activation, sustained time per channel, and inter-channel activation intervals. These help distinguish short taps, sustained touches, and continuous motion. 2) Spatial features: including the number of activated channels, their distribution, and activation sequence. These help classify localized contact, broad-area touch, and directional motion. Using these features, multi-channel signals are organized into distinguishable gesture samples.

For data collection, three participants were invited to perform each gesture 100 times, yielding 1500 samples. Signals from all 12 channels were recorded synchronously, segmented, and transformed into feature vectors for training and analysis.

For gesture classification, the extracted spatiotemporal features were used to train an SVM classifier with a radial basis function (RBF) kernel (see Figure 3d). The dataset was evaluated using five-fold cross-validation. On the five-class

Table 1. Five Natural Tactile Gestures and Their Spatiotemporal Features

Gesture	Temporal Features	Spatial Features
Down-stroke	Sequential activation, reverse direction	Adjacent units activated downward
Up-stroke	Sequential activation, clear direction	Adjacent units activated upward
Poke	Short duration	Small area, single-unit activation
Pat	Short duration	Large area, multi-unit simultaneous activation
Press	Long duration	Large area, multi-unit simultaneous activation

gesture recognition task, the model achieved an overall accuracy of 87.5%, with an average F1-score of 0.86. Confusions mainly occurred between pat and press due to their similar large-area activation patterns, whereas directional stroking gestures were more reliably distinguished by channel activation order. These results indicate that the proposed method can reliably distinguish natural tactile gestures for real-time robot feedback control.

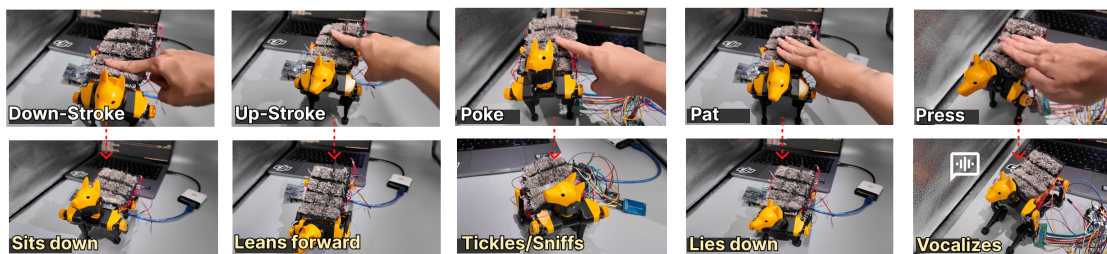
5 Plush Skin-Based Robotic Dog Interaction Demo

After achieving recognition of natural tactile gestures, we mapped them to responsive behaviors of the robotic dog, constructing a natural interaction demo. The plush skin serves not only as an input layer but also as an interactive interface connecting user touch and embodied robot response.

The mapping between gestures and robot feedback is summarized in Table 2. It was designed with consideration of the affective and relational qualities associated with different forms of touch. This mapping transforms natural touch into an interpretable interactive vocabulary, and the robot’s responses form embodied replies.

Table 2. Interaction Semantics and Feedback Mapping of Natural Tactile Gestures

Gesture	Interaction Semantics	Robot Feedback
Down-stroke	Affectionate, calming	Sits down
Up-stroke	Guiding, inviting	Leans forward
Poke	Playful	Tickles/Sniffs
Pat	Alerting, attention-seeking	Lies down
Press	Deliberate input	Vocalizes


 Fig. 4. Interaction between the user and the robot dog through *Pluskin*

In terms of control implementation, the plush skin captures tactile input, the recognition module classifies the gestures, and a host computer maps the results to control commands sent to the *Petoi Bittle*, which executes the

corresponding behavior. This closed-loop interaction flow enables the plush skin to facilitate real-time interactive control.

During the demonstration, users engage with the robot through a low-barrier interface. By emulating natural human-animal bonding, users interact intuitively by touching the plush surface as they would a real pet, eliciting varied responses as illustrated in Figure 4. By transcending conventional modalities such as buttons, screens, or voice commands, this approach leverages bodily contact to foster a more immersive, natural, and emotionally resonant human-robot companionship.

6 Discussion and Future Work

This research establishes *Pluskin*, not merely as a sensing device, but as an affective gesture interaction interface for companion robotics. By transforming plush material from a passive decorative layer into an active, perceptive robotic skin, we bridge the gap between tactile softness and functional intelligence. This dual-role architecture allows the robotic surface to serve simultaneously as an inviting tactile medium and a tactile input interface, effectively turning the robot’s entire body into a front-end channel for human-pet communication. This shift from decorative to functional materiality presents the textile skin of a companion robot as a site of active engagement rather than a static shell.

Beyond the technical fabrication, this work underscores a transition from imperative to affective interaction paradigms. While traditional input modalities like buttons or voice commands rely on discrete, command-execution logic, *Pluskin* facilitates a contact-response loop that aligns with the inherent materiality of plush surfaces. By interpreting natural and non-imperative gestures, such as stroking, patting, and pressing, the system moves away from cold, binary instructions toward a more intuitive and emotionally resonant language of touch. This approach acknowledges that in the context of companionship, touch is not merely a data input but a vital mode of relational signaling, fostering a more immersive and low-barrier form of human-robot bonding.

The current system is limited to five predefined gestures and a 4×3 sensing array, and the interaction demo is implemented on a single robotic platform. Future work will explore a larger gesture vocabulary, higher spatial resolution, and richer forms of robot feedback to support more expressive tactile interaction.

7 Conclusion

This research presents *Pluskin*, a perceptive skin system that transforms traditionally decorative plush into a functional tactile interface. The core innovation lies in the single-step digital embroidery of a thick, loop-pile fibrous structure, which provides the volumetric compliance necessary to modulate electrical resistance via the contact piezoresistive effect. By engineering the textile’s physical morphology into a sensing medium, we facilitate a shift in human-robot interaction that replaces discrete, imperative commands with a continuous, affective contact-response loop.

The technical viability of this paradigm is validated through a 4×3 sensing array and a spatiotemporal machine learning pipeline integrated into a robotic dog. By accurately classifying five categories of natural social gestures—ranging from rhythmic stroking to impulsive patting—the system demonstrates that materiality-driven tactile perception can support more intuitive and emotionally resonant companionship. Ultimately, this work not only introduces a novel fabrication method for functional soft surfaces but also underscores the value of synergizing textile materiality with tactile intelligence to create more affective and natural human-robot relationships.

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